

TDR4M1

Swerve Drive Design Report

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Project Calender

Sun	Mon	Tue	Wed	Thu	Fri	Sat
			1 - Robot Kit	2 - Identify Situation and problem & Research	3 - Fusion 360 Test	4
5	6 - Research	7 - Research/Concept drawings	8 - Build Kitbot	9 - Build Kitbot	10 HOLIDAY	11
12	13 HOLIDAY	14 - Cad	15 - Cad	16 - Cad	17 - Cad	18
19	20 - Cad/Print	21 - Fabrication	22 - Fabrication	23 - Fabrication	24 - Fabrication	25
26	27 - Evaluate	28 - Redesign	29 - Redesign	30 - Redesign	31 - Make Second Module	1
2	3 - Redesign	4 - Redesign	5 - Redesign	6 - Redesign	7 - Final Due Date	

S.P.I.C.E Design Report

Situation

The Iroquois Ridge class of 2026 have been tasked to create a swivel drive mechanism to construct a drivetrain. To start, VEX motors are provided with 200rpm cartridges each. For wheels we are limited to 4 inch omni and traction wheels that are contained within the VEX classroom kits. The swivel drive system is required to reach target speed of 6.5 feet per second with the given 4 inch wheels.

Problem

The task assigned is to create a robust and efficient robot designed in Fusion 360 that uses a swerve drive chassis. This will involve creating parts and assembling them for the drivetrain, gearing and full assembly. Usable parts are limited to what is provided in class from VEX V5 parts and 3d printed parts.

Additionally, we will need to calculate the gear ratio required to achieve the maximum target velocity of 6.5 feet per second, given that we use 600 rpm motors with 4 inch wheels.

It must also be mounted to ensure effective operation even with the 4 motor limit. The 4 motor limit means we can only have two drive modules and two castor wheels. The two castor wheels need to be designed in order to not inhibit the maneuverability of the robot.

Due to the nature of the project being mainly 3d printed, tolerances are a big factor to consider. Shaft clearances must be tested for an ideal fit, heat set inserts tightness, gear dimensions and the mesh between them. Another issue that needed consideration is that printing out of standard PLA means large plates will flex, leading to gears momentarily losing contact and skipping.

An encoder was determined to be required to get the rotation of the wheel hub for an absolute reading on rotation compared to using the built in motor encoders which lose accuracy if gears skip.

A cross-brace was determined to be required due to the wobbly structure of the overall frame which inhibited all four wheels from having consistent contact with the ground.

Another point to consider is the height of the robot, since the wheel is 4", after building the mechanism, we must ensure that the center of gravity is still low so that sudden movement will not tip the robot.

Investigation

Relevant terminology:

Swerve drive - *Another word for swivel drive*

Gear ratio calculation

Motor RPM - 100/200/**600**

Target robot speed - 6.5 feet per second / 78 inch per second

Wheel diameter - 4 inch

Wheel circumference = $4 * \pi = 12.5663706$

Target wheel RPM = $78 / 12.5663706 * 60 = 372.423$ RPM

Target overall gear ratio with 600 = $372.423 / 600 = 0.620705$

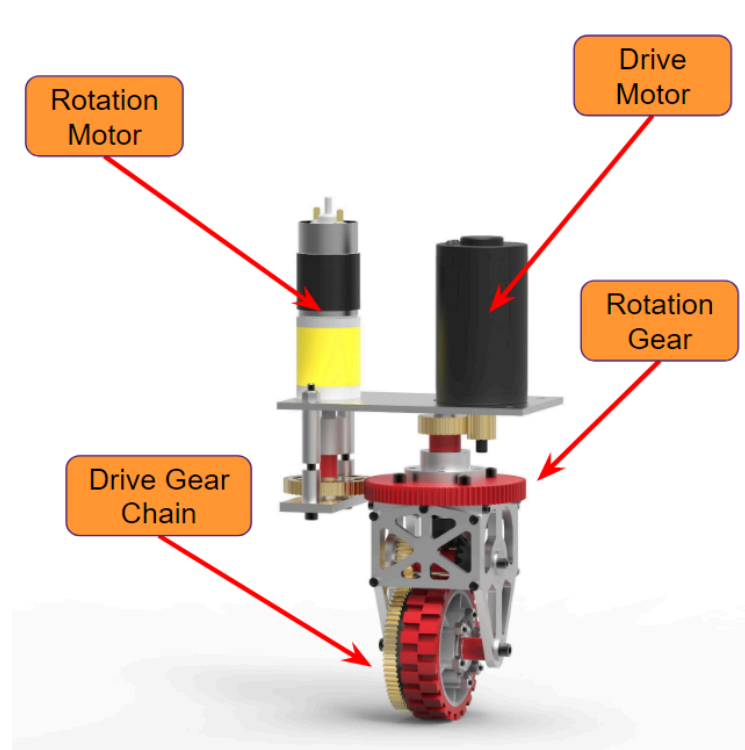
Therefore the required overall gear ratio is approximately **1 : 0.620705** with 600 cartridges

Thus we will use a gearing down with a ratio of 2:3 on 4" wheels with a 600rpm cartridge.

Research

Concept #1 Swerve Drive

Coaxial Swerve Mechanism



Applicability

This concept has a separate rotation motor and drive motor. It allows for full 360 degree rotation of the wheel. This application would be viable in our situation as it doesn't require complex gear geometry and meets the motor requirements.

Pros	Cons
- Allows for unlimited 360 degree rotation of wheel - Can be made using COTS vex gears - Easy to mount encoder - Simple mounting system, low gear quantity required, thus less velocity lost to friction - Compact design	- Only has one motor driving the wheel, makes less effective use of the 4 max motors - The motor sticks out on top, limiting mounting space on top

Concept #2 Swerve Drive

Differential Swerve Mechanism



Applicability

This design is impractical for our situation. It requires complex custom gears with a hole in the middle and bevel gears alongside special bearings to mount the gears which we do not have.

Pros	Cons
- Uses two motors to turn and drive the wheel, allowing for full use of power from all four motors to drive the wheel - Wheel being partly in the gear allows for a much more compact design - Allows for unlimited 360 degree rotation	- Requires custom gears - Difficult to design - Special V profile bearings are needed mount the gears around the wheel which aren't available to us - Relies heavily on bevel gears which are hard to mount as they need to be strongly supported on both ends - Difficult to mount encoder

Concept #3 Swerve Drive

Alternate Differential Swerve Mechanism



Applicability

This design is impractical for our situation as it requires belts which we don't have access to and generate excessive friction.

Pros	Cons
- Uses two motors to turn and drive the wheel, allowing for full use of power from all four motors to drive the wheel - Easy to mount encoder - Allows for unlimited 360 degree rotation	- Required custom printed gears that aren't complex - More complex gearing - Requires belts which we don't have access to

Concept #4 Swerve Drive

Non-Coaxial Swerve Mechanism

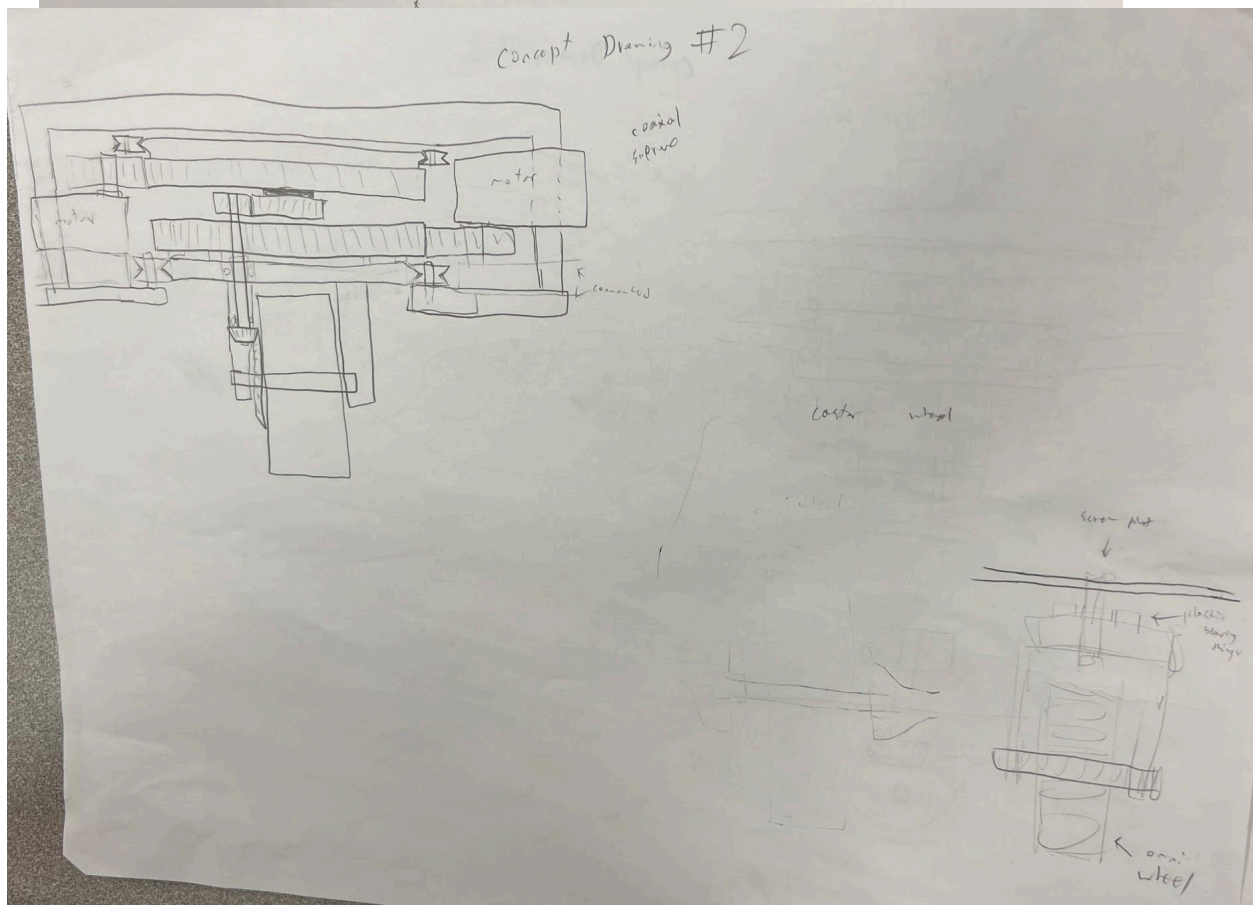
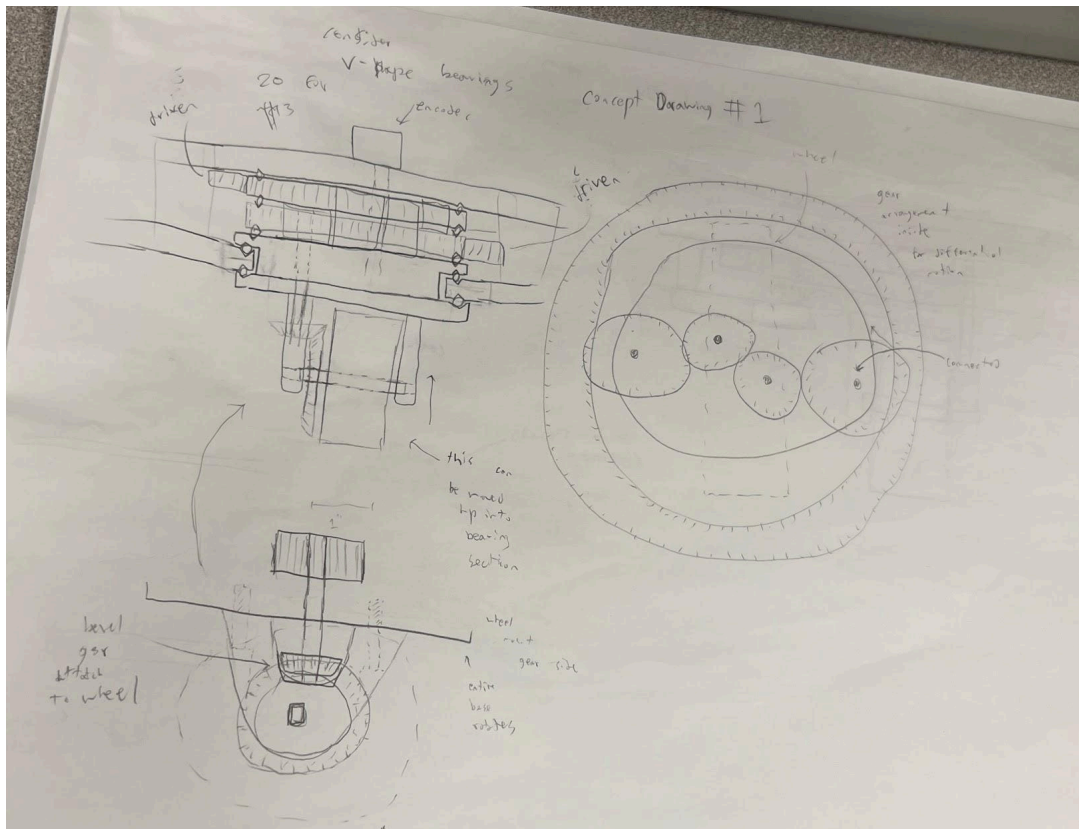


Applicability

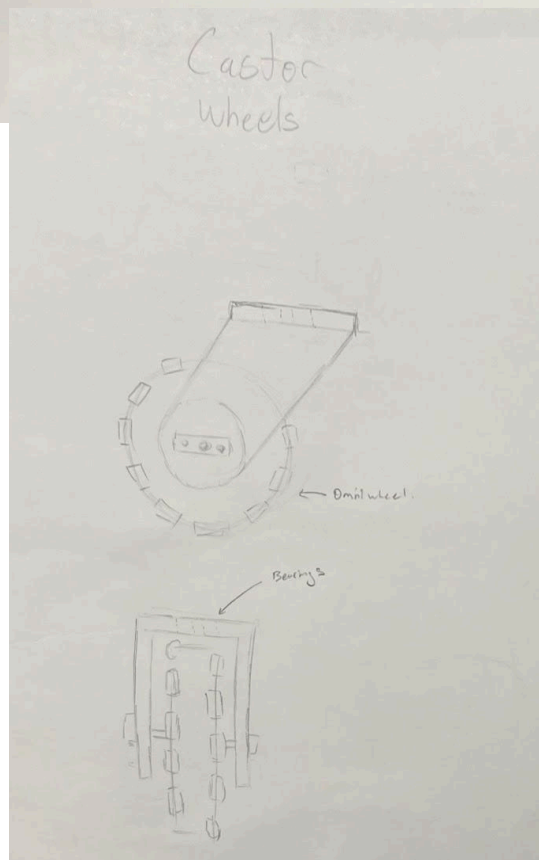
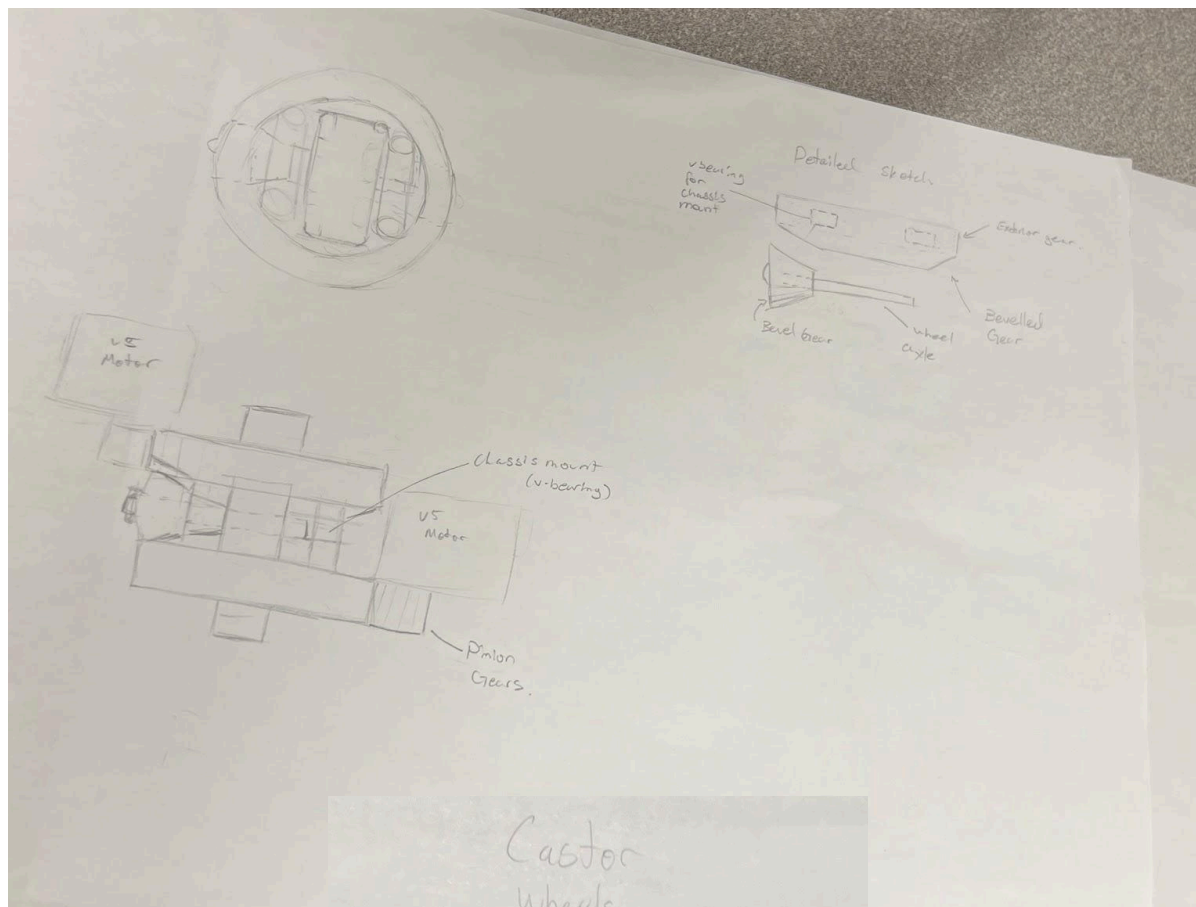
This design is great in theory, however it has the one main limitation that the swerve wheel cannot rotate 360 fully since the motor wire will be a hard limit. Chaining the drive motor also creates an excessive amount of friction, slowing us down.

Pros	Cons
- Uses a motor each to drive and swerve - Easy to mount encoder - Low gear usage	- Chain required to power wheel, this generates a lot of friction from slack. - Restricted range of motion due to motor spinning with wire. - Since the design requires the motor to be on top of the wheel this takes up a lot of precious y axis space.

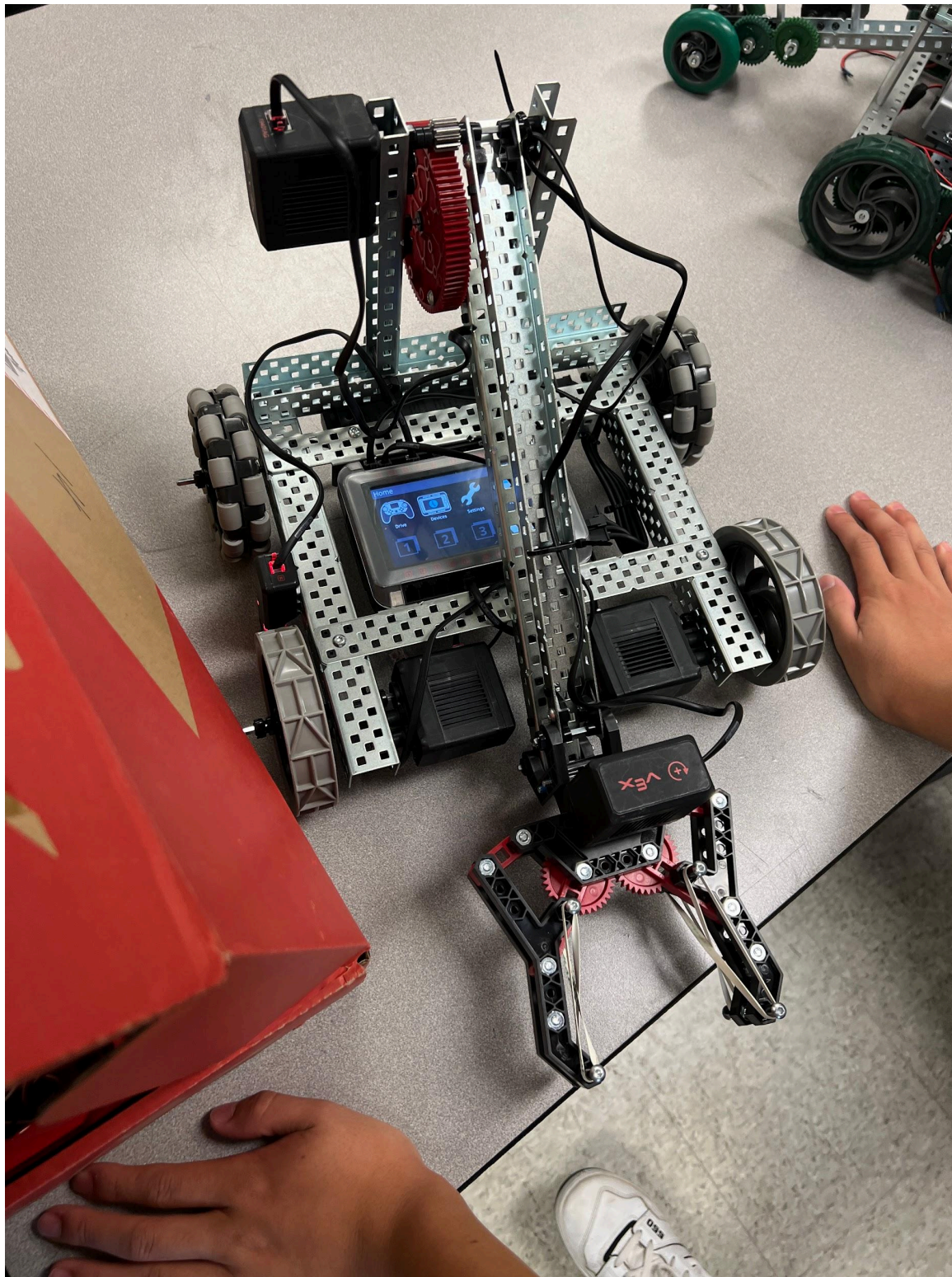
Concept Designs



Final Concept Sketch



Construction



Assembled kitbot (Oct 22)

Relevant Information

- V-bearing amazon link along with specifications
 - [Here](#)
- **Gears**
 - Pitch 16 tpi
 - Pressure angle 20

Gear Ratio Calculations

Motor RPM - **600**

Target RPM with 4" wheel diameter - **372**

Target reduction - **1 : 0.620705**

Calculations in Investigation Section

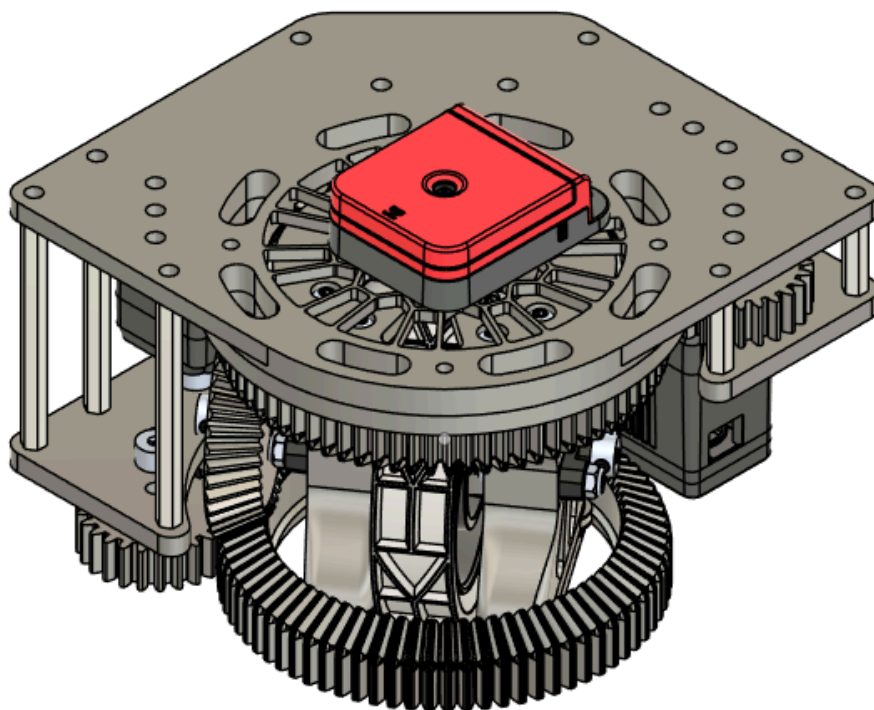
	Ratio one	Ratio two	Final Ratio
Driving	25	75	.625
Driven	75	40	Final RPM
Ratio	0.33333333	1.875	375

Swerve Drive CAD design iterations

Swerve Drive iteration one

Potential Concerns

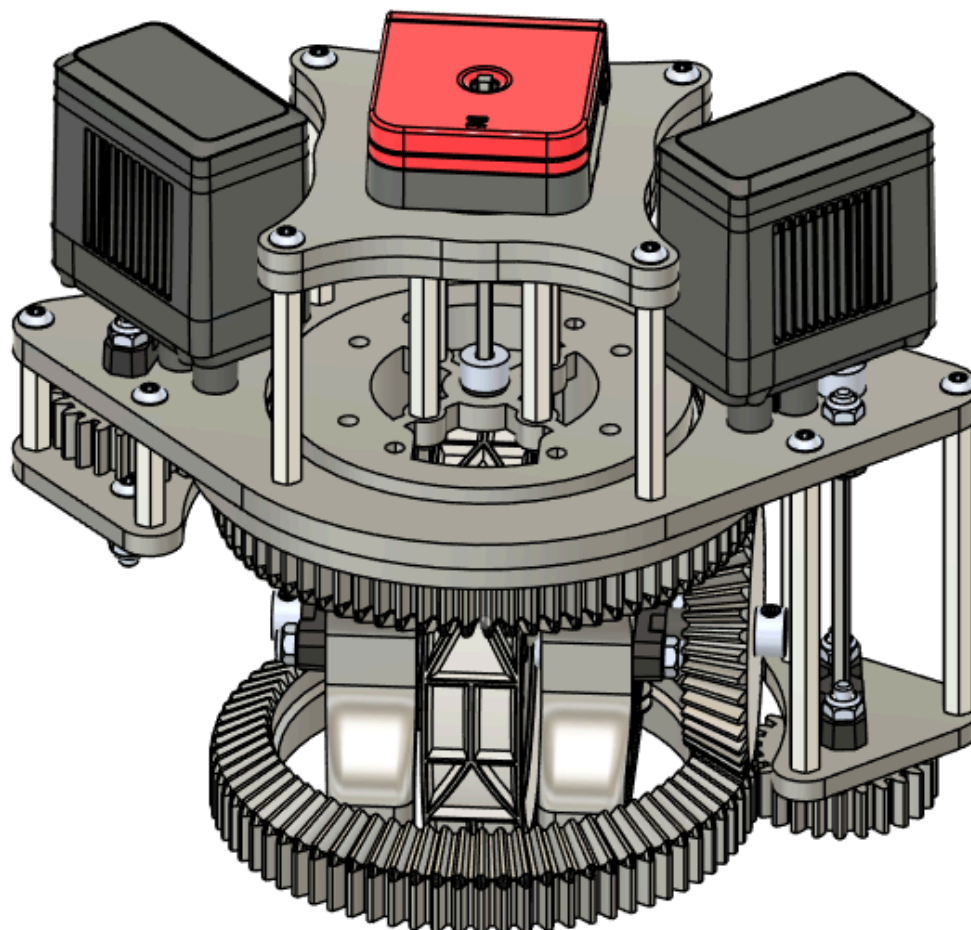
Very large surface area, approximately, 7 x 7 inches. Excess of wasted space within the 7x7 plane, and the overall volume makes the robot very bulky.



Swerve Drive iteration two

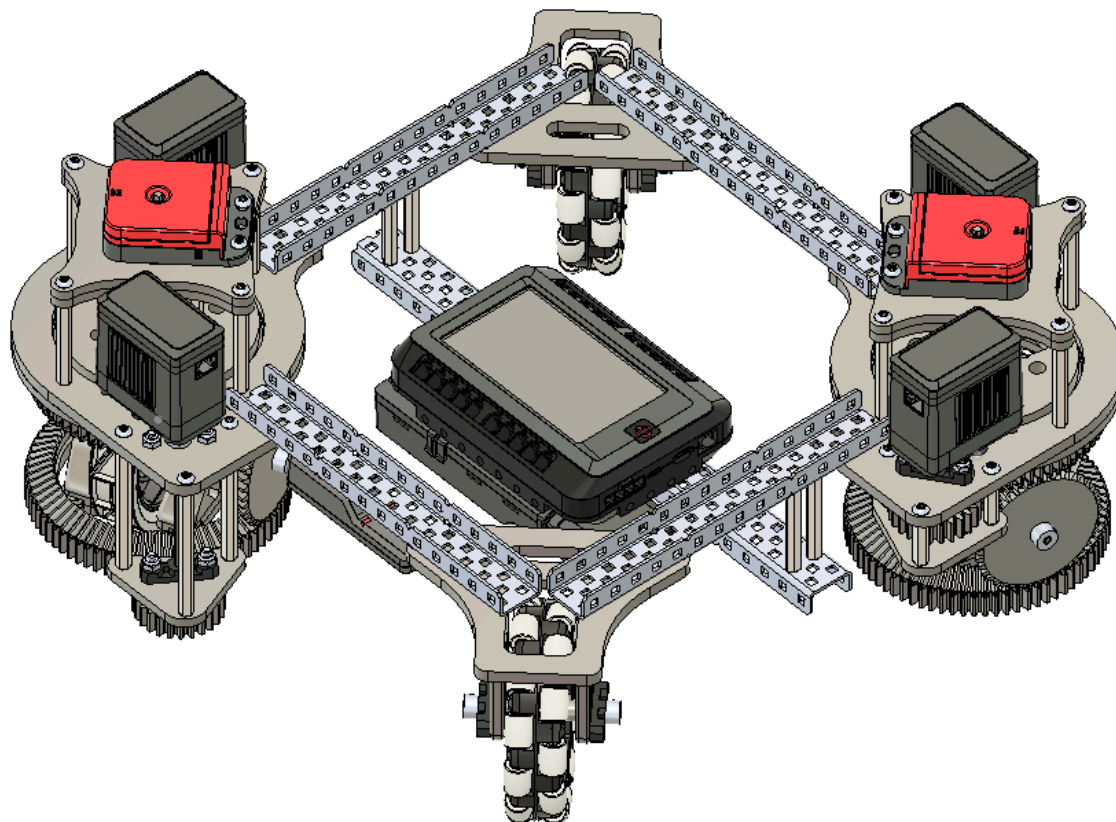
Potential Concerns

Lower pinion gear may shift under constant load due to standoff loosening. This may bend the entire triangular piece away from contacting the main gear, losing power or even connection.



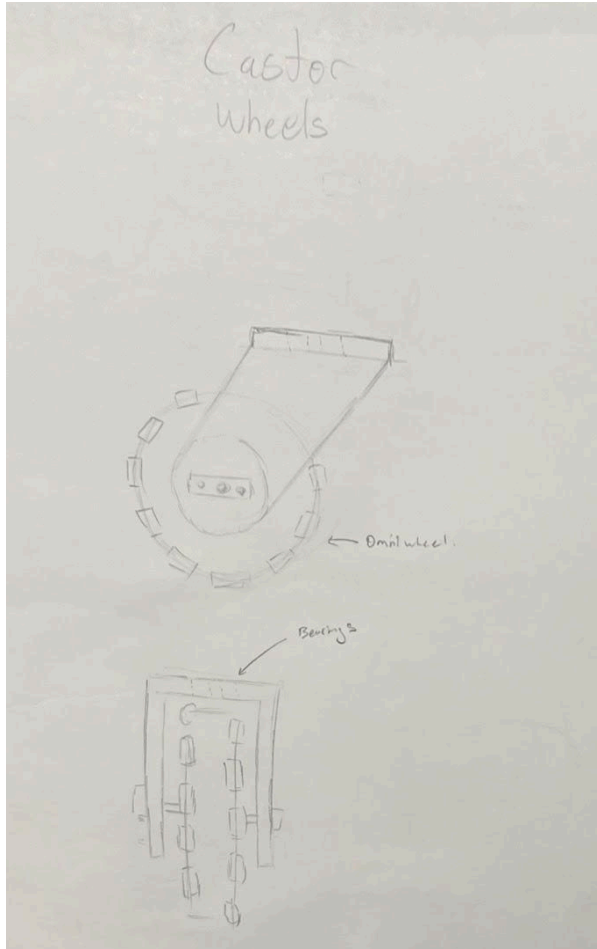
Full Chassis Cad

Includes one cross brace to improve rigidity and allows lower mounting of brain and battery to lower center of gravity. (Oct 30) (Below)



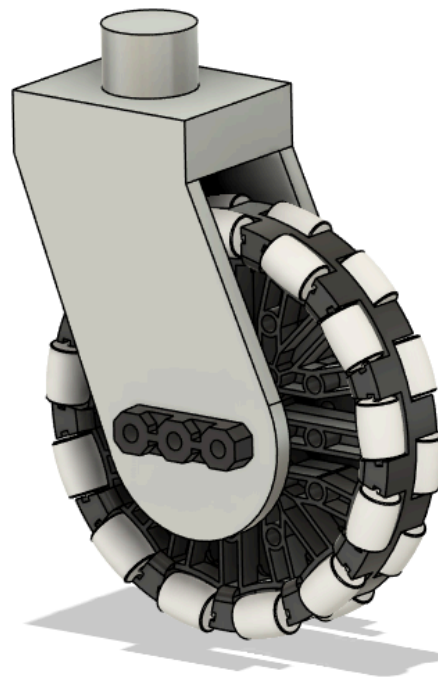
Castor Omni Wheel CAD design Iterations

Castor Omni Wheel Iteration one

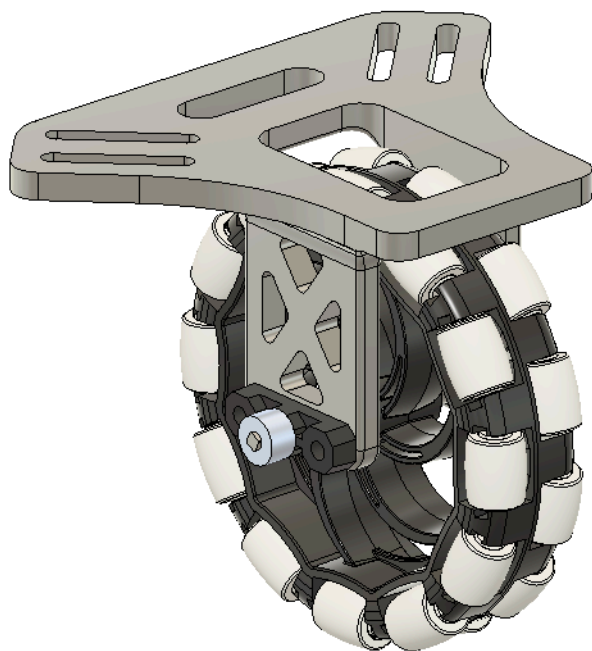


The first iteration of the castor wheel was designed to be swivel without the intervention of a motor. The design was planned to have a v-bearing that allowed for 360° rotation with minimal friction. (Oct 28)(Left, Bottom)

This design has one main flaw, that is even more pronounced by the 4" wheels at our disposal. For a castor wheel to turn, the wheel must travel an arc to reorient itself in a new direction. Due to the large wheel, the arc would significantly reduce the robot's ability to make tight turns.



Iteration Two



However, this idea was later modified to have a stationary omni wheel, providing all the same benefits for little to no compromise. One piece omni caster wheel, fixed at a 45 degree angle. The Omni wheel allows it to move in any direction without excessive friction. (Oct 29) (Left)

Caster Omni Wheels fully assembled, worked effectively and the wheel was held in place with sufficient rigidity. Force was applied onto the mount to mimic the load of the robot. Minimal bending occurred and the part was pronounced complete. (Oct 30) (Bottom)



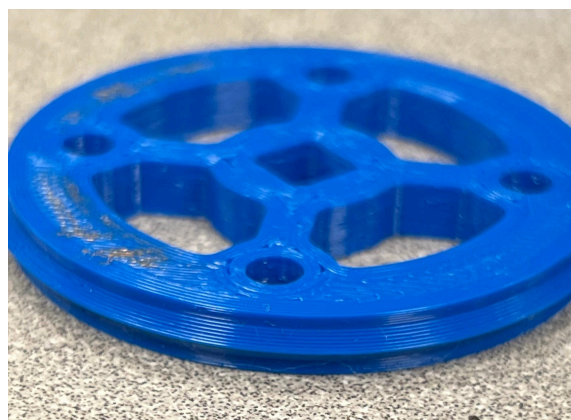
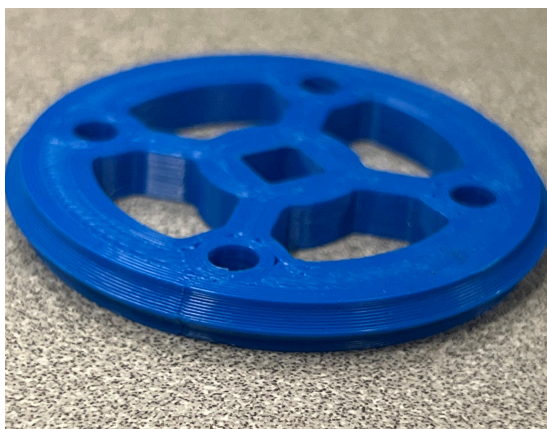
Swerve Drive Print Tolerances



Tolerance test for Inserts, determined to be 0.003" on each side (Oct 23) (Below)

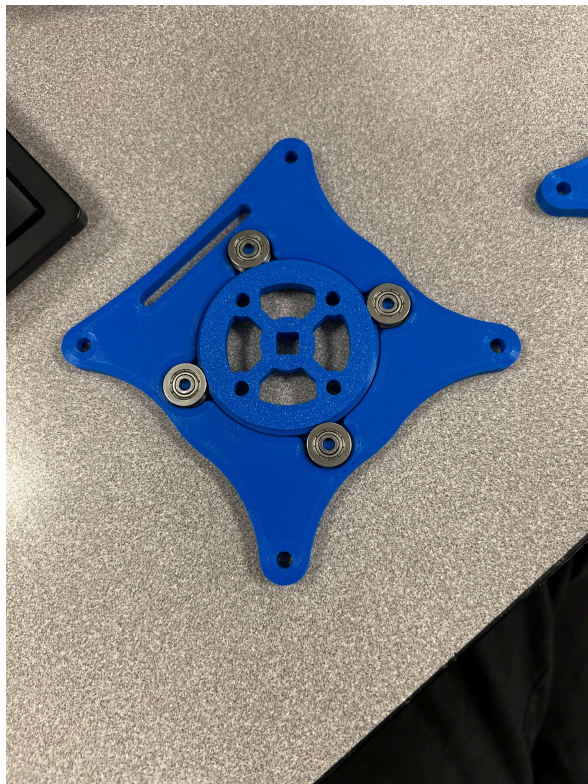


Tolerance test for bearings, 0 clearance works (Oct 23) (Above, Left)

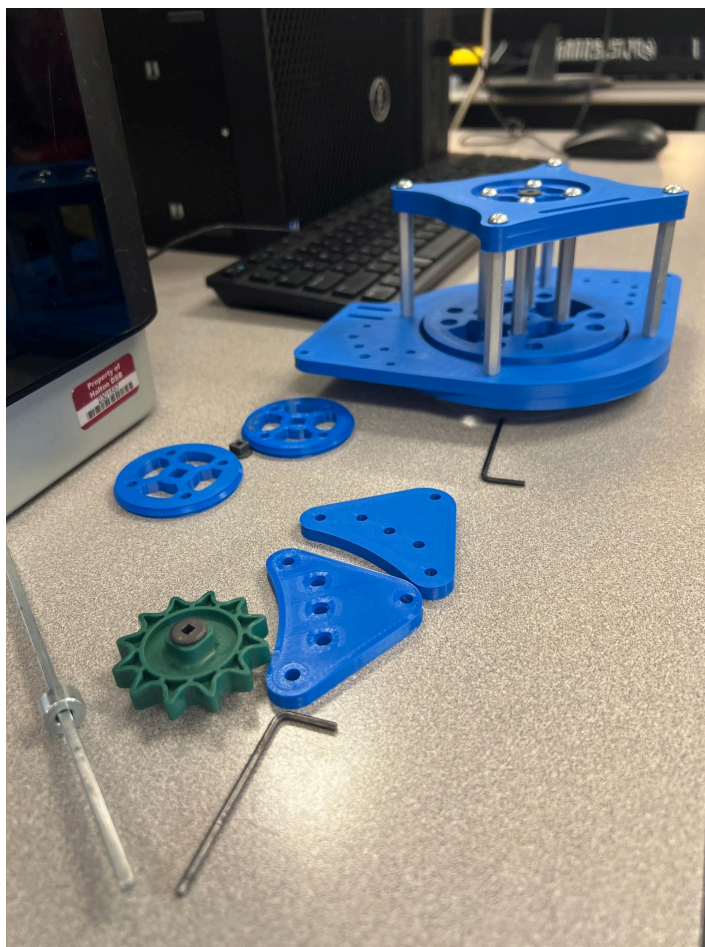


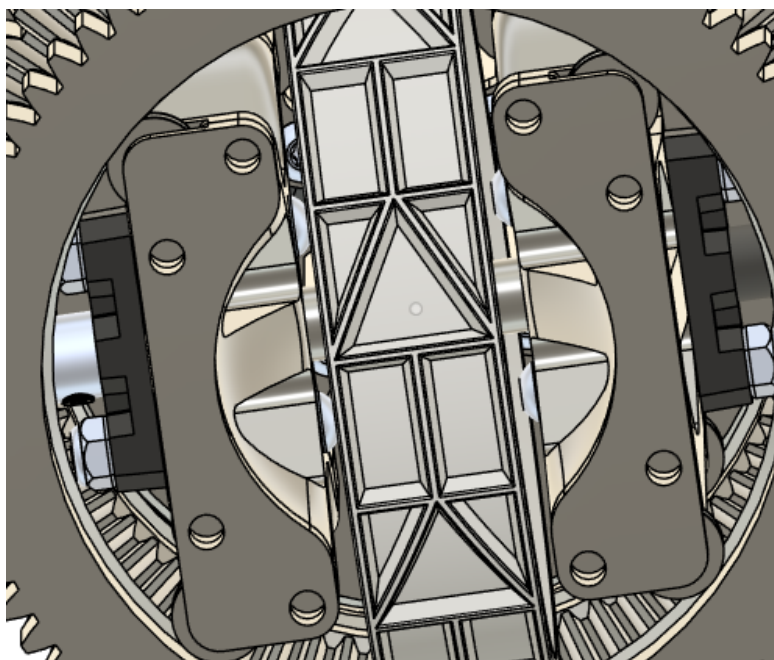
Test V-rings printed, clearance was found to be 0.001" for a snug fit. Seam arrangement changed to random because as seen on the left V-ring, the seams form a line which causes a noticeable bump while spinning. (Oct 24) (Above)

Upper V-ring holder partly assembled with bearings.
(Oct 24) (Left)



Main upper mount printed, assembled and tested. The biggest flat plate was slightly warped due to pulling it off the print bed too fast. (Oct 24) (Right)



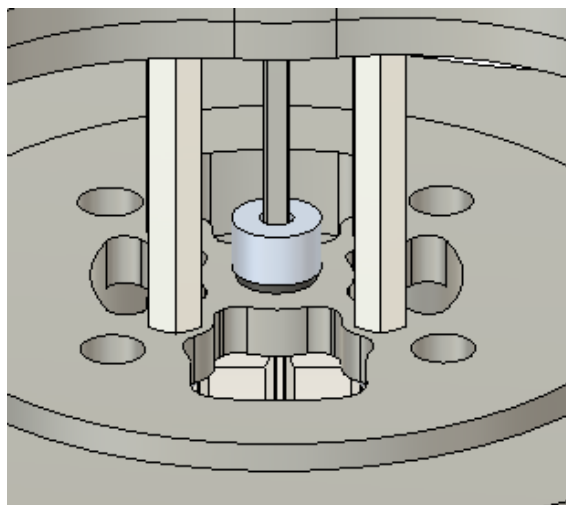


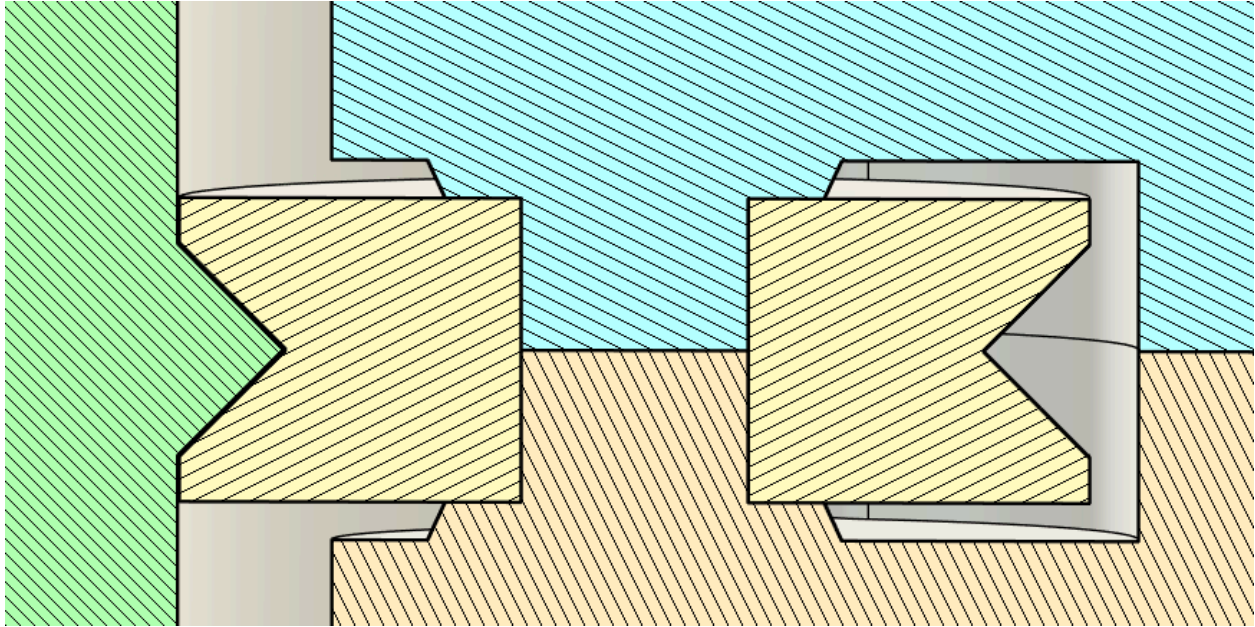
CAD parts for wheel hub assembly Along with the bearing mount. (Oct 24) (Left)



2nd Iteration

The two vertical support columns initially were CADded to contain 8 heat set inserts for each due to the purpose of stability, however it was found that 4 on each was perfectly fine. CAD parts for wheel hub assembly as well as reprinted large flat plate so it is now full flat. Heat set inserts were inserted into the wheel hub mounts. (Oct 27) (Left and Under)



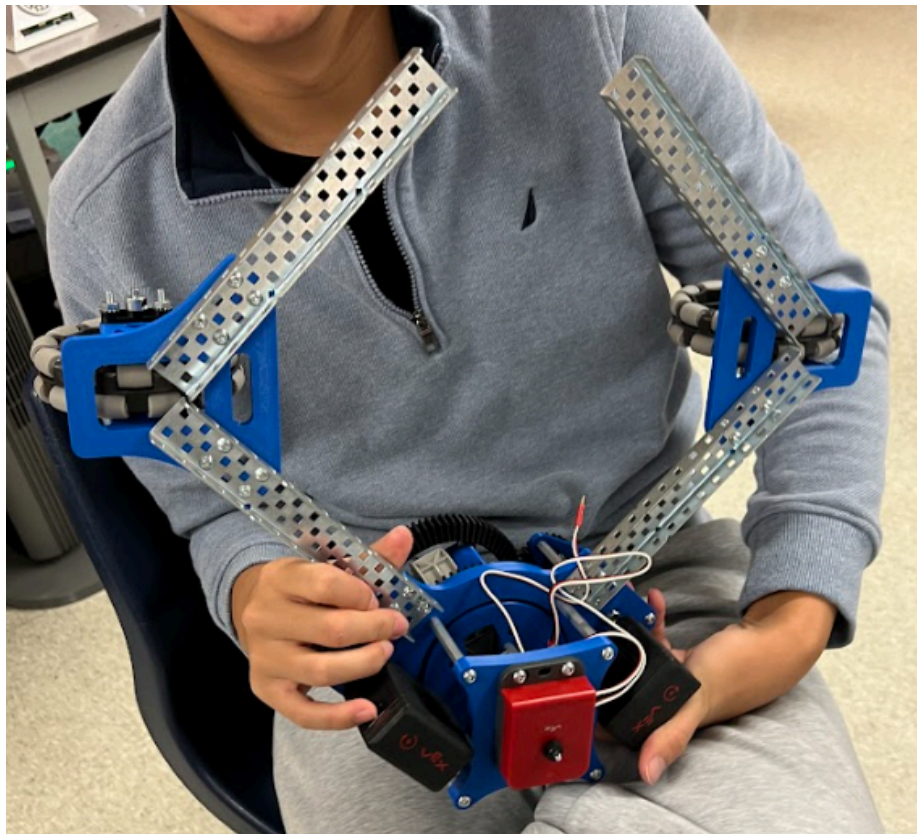
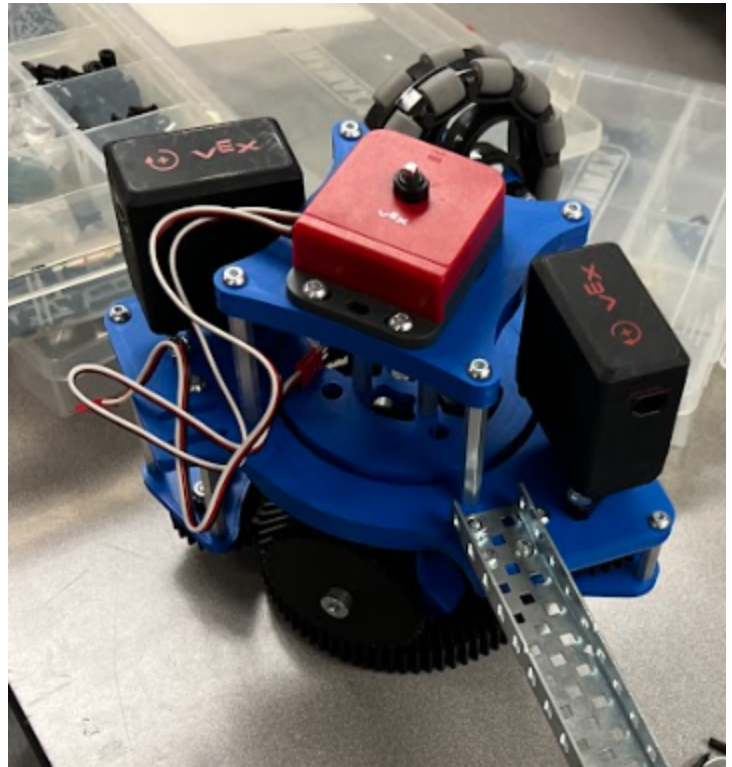


(Top) Bearing: Yellow, Gear: Green, Wheel mount: Blue/Orange. Cross section of a sample bearing mount, specifically the bottom differential gear. As seen in the image, bearings are lifted off of both wheel mount surfaces to reduce friction.



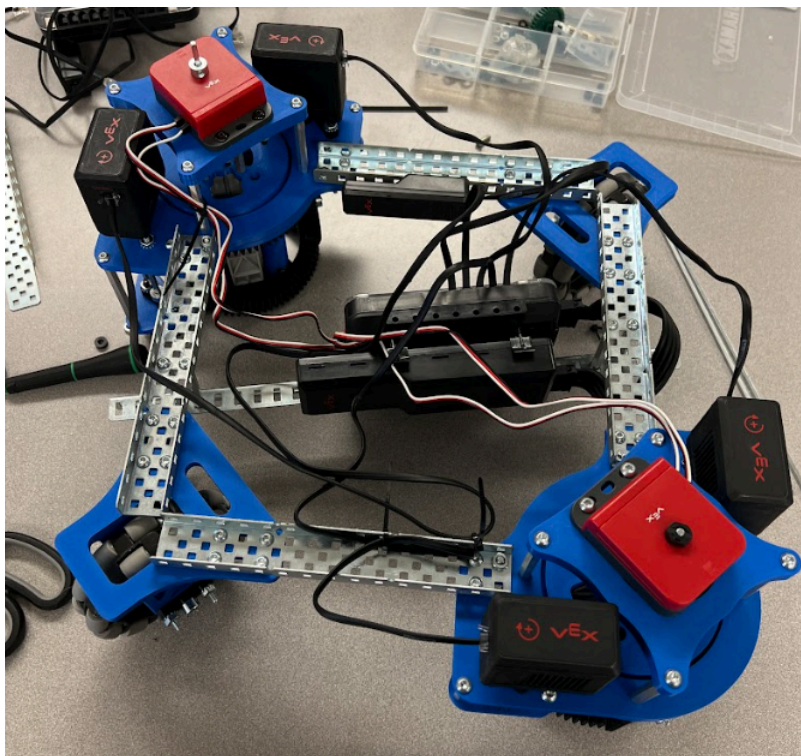
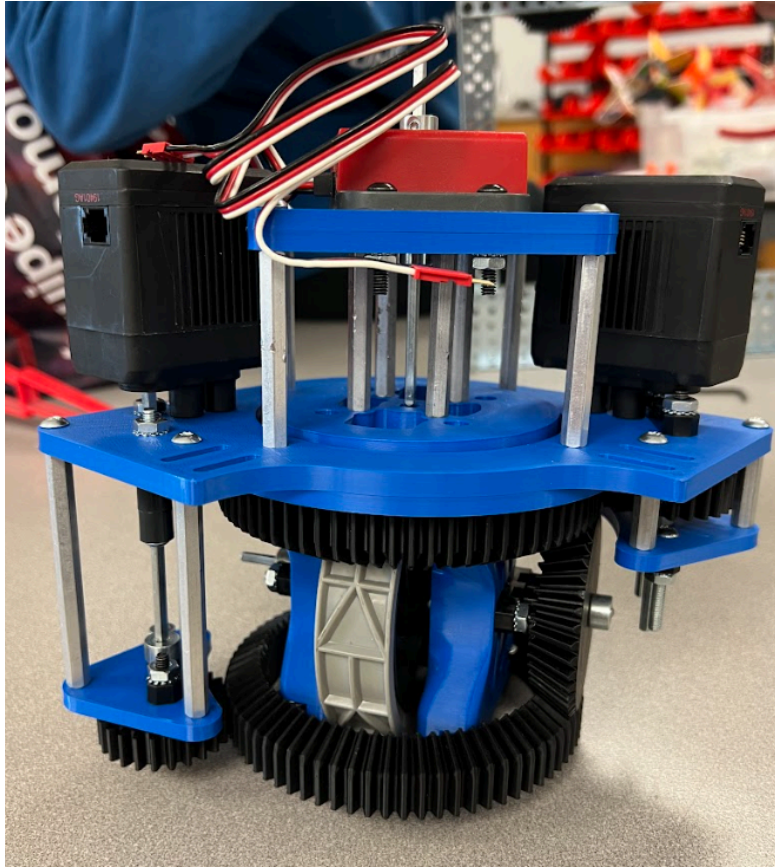
Initial wheel hub fully assembled and spun. An acceptable amount of friction was found and bearings held gears in place with enough rigidity to maintain full contact at all times. (Oct 27) (Left)

The wheel hub, main body of the swerve module, motors and encoder were assembled. The full module was powered and tested, no issues were found. (Oct 30) (Right)



3 corners of chassis fully assembled and connected. Was slightly wobbly and required a cross brace. (Oct 31) (Left)

Second Swerve Drive Module
Second module fully assembled
and tested. (Nov 5) (Right)



**Full Chassis Assembled,
Wired, and Tested (Nov 6)**
(Left)

Evaluation

Individual Swerve Module

The design achieved an approximate top speed of 7.0ft/s, measured using a timed distance test. This exceeds the planned goal of 6.5ft/s by 12% under the same 4 VEX V5 11W motor limit. It also demonstrated exceptional maneuverability, being able to effortlessly “swerve” across the floor.

The design had exceptional rigidity and structural integrity even while powered and spinning. This is primarily due to the V-groove bearings that allowed for the rigid yet low friction mounting of the two main gears and the entire wheel hub. The bearings also functioned, albeit at increased friction, while a load (~20 pounds) was applied directly to the main plate of the module. One issue with these V-groove bearings is that the upper main gear experienced noticeably more friction than the lower main gear. This can be attributed to how close it is to the plate it is mounted to as it is unable to flex inwards whereas the lower main gear is able to slightly flex inwards which seats the V-groove bearings to exert a perfect amount of pressure onto the lower main gear. The upper gear being unable to flex inwards is pressed into the upper main gear extremely tight, pushing the bearing to its radial load capacity and therefore increasing friction in the system.

Tolerance tests done for inserts allowed for all of the inserts used around the module to fit perfectly on the first attempt, making for a smooth assembly process.

Additionally, with motors mounted at the top of the main plate, it means it will have a high center of gravity that may cause the robot to tip when making sharp maneuvers. The motors could have been mounted below but would have resulted in a significantly larger silhouette and more complex mounting plates.

Omni Castor Wheel

The Omni Castor Wheel proved to be extremely effective and simple to assemble. It consisted of one main plate and 2 custom sized spacers all to be printed and utilized regular vex bearings, screws, and nuts to fasten it to the frame. The large fillets on the main plate wheel hub also held the wheel in place very well.

Full Swerve Chassis

The overall function of the swerve chassis was sub-par, although it could move and effectively “swerve”, the wheels often went out of sync with one another and would cause it to stop driving properly. This issue is a programming error as it would lose the rotational position of the wheel from the encoder and fall out of sync.

Another problem encountered was the lower drive gear on both modules was frequently skipping under significant changes in force due to deflection in both the mounting plate of the lower pinion gear and the main wheel hub. A band-aid fix to this issue was adding rubber bands to pull the mounting plate of the lower pinion gear towards the main gear and keeping the two gears in contact at all times. This problem with the pinion could be fully fixed by adding more

reinforcement to the main plate it is mounted to with rigids or by making it thicker to reduce deflection. Deflection in the main wheel hub can be reduced with tighter tolerances on the V-bearing rings that may come at the tradeoff of increased friction when rotating.

The full chassis initially suffered from a lack of rigidity but adding a simple cross-brace significantly improved the strength of the frame. The cross-brace also acted as an effective mounting point for the brain and battery at a low and center point on the robot.

The castor wheels mounted to the chassis worked extremely well to keep the robot stable on the ground and did not inhibit the movement of the robot with excess friction.